



1st DDP Review

on

Experimental Investigation on Unsteadiness in Crossing Shockwave Boundary Layer Interaction

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Introduction and Motivation

Introduction:

- Crossing shockwave boundary layer interactions (CSBLI) occur when two oblique shocks generated by a double fin set-up interact with a turbulent boundary layer.
- The interaction is highly three dimensional, often involving boundary layer separation and vortex structures.

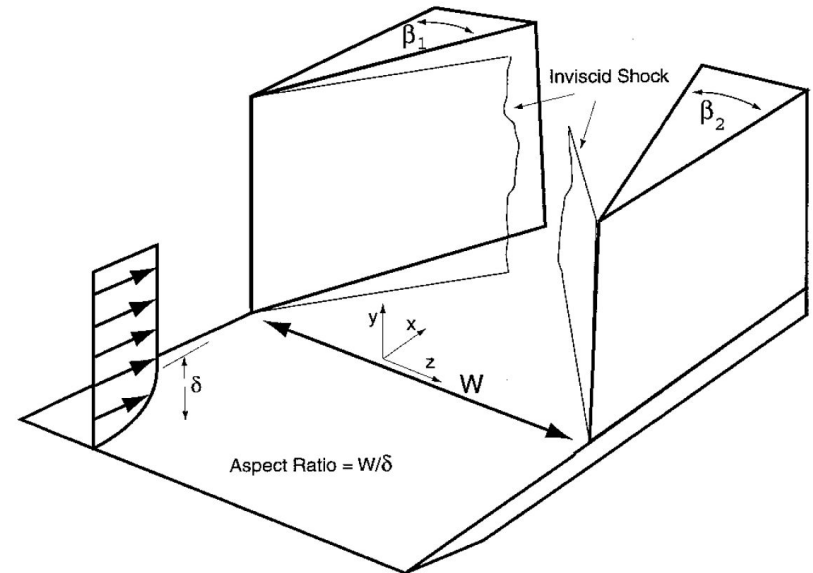


Fig1: Schematic of the double-fin generated crossing shock-wave/turbulent boundary-layer interaction [1]

Motivation:

- Unsteadiness in shock interactions and shock boundary layer interactions have been extensively investigated, through dominant frequency modes and in multiple configurations spanning from the flow over ramps and around three dimensional protuberances.
- However, the associated unsteady behaviour, MR-RR transition and associated hysteresis in CSBLI have been relatively unexplored, motivating the present investigation.



Literature Review

- The low-frequency unsteadiness associated with two dimensional SBLI have been widely studied and understood. Due to the multiple qualitative differences in the structure of both these interactions, Gaitonde and Adler (2023) [2] argued that the mechanism governing the unsteadiness in 2D SBLI may not be applicable for the 3D case.
- In swept ramp induced three-dimensional SBLI, the shock oscillations were found to be of mid frequencies with smaller amplitudes. Adler and Gaitonde (2020) [3] found the low-frequency oscillations associated with 2D interactions to be significantly suppressed.
- Bow shock induced 3D SBLI were found to exhibit low-frequency oscillations by Ramachandra et al. (2023) [4]. Vayala et al. (2026) [5] attributed this unsteadiness to an imbalance and time lag between mass injection at reattachment and mass escaping spanwise through the horseshoe vortex.
- Crossing shock wave boundary layer interactions (CSBLI) have also been observed to exhibit unsteadiness. Poddar and Bogdonoff (1990) [6] reported high-frequency large amplitude disturbances downstream of the shock-crossing location.
- Davis and Hingst (1992) [7] identified dominant fluctuations around 1 kHz but also high-amplitude unsteadiness near 150 Hz. Seçkin et al. (2023) [8] also reported significant low-frequency unsteadiness. Despite these observations, the mechanisms governing this unsteadiness in CSBLI remain unaddressed.



Experimental Methodology

- Experiments will be performed in the Mach 2.87 supersonic wind tunnel at the Gas Dynamics Laboratory.

Conventional Schlieren and Oil Flow Visualisation:

- Fins mounted on the vertical acrylic side wall, allowing conventional schlieren to directly image the foot of the shock interaction and capture its unsteadiness. Oil flow visualization will also be performed on the same configuration.

Unsteady Pressure Measurement:

- Fins mounted on the tunnel floor, unsteady pressure transducers used to measure the fluctuating surface pressure field along the centerline. Oil flow visualization will be performed for this configuration as well.

Tomographic Background Oriented Schlieren:

- Tomo-BOS will also be employed to obtain a three-dimensional visualization of the flow structures, overcoming the line-of-sight integration inherent to traditional schlieren imaging.



Data Analysis

Post Processing Methods

- **Power Spectral Density:** to identify dominant frequencies of shock/separation unsteadiness from pressure data
- **Proper Orthogonal Decomposition (POD) and Spectral-POD:** to extract dominant modes and their associated frequencies from schlieren images.
- **Coherence and cross-correlation:** to establish spatial relationships and phase lags between locations on the centreline, possibly revealing the governing mechanism of unsteadiness.

MR–RR transition and Hysteresis :

- Crossing shocks can settle into Regular Reflection (RR) or Mach Reflection (MR), with a dual-solution domain.
- Variable fin angles will be adopted to approach the dual-solution domain from both increasing and decreasing deflection to test for hysteresis.
- The effect of MR-RR transition on the unsteadiness will also be examined.

Work done during Summer

Calibration of BOS using an Axisymmetric Open Jet

- Open jet flow from an axisymmetric nozzle was captured using Background Oriented Schlieren. The motion of the dots on the speckle was used to find the associated displacements using an open source software ComBOS developed by Mercier and Lacassagne (2023) [9].
- The data was then post-processed to reconstruct the density field, using Abel Inversion and Filtered Back Projection, similar to the procedure used by Venkatakrishnan et al. (2004) [10]
- The results gave some insight on the feasibility of using tomo-BOS for 3D visualization.

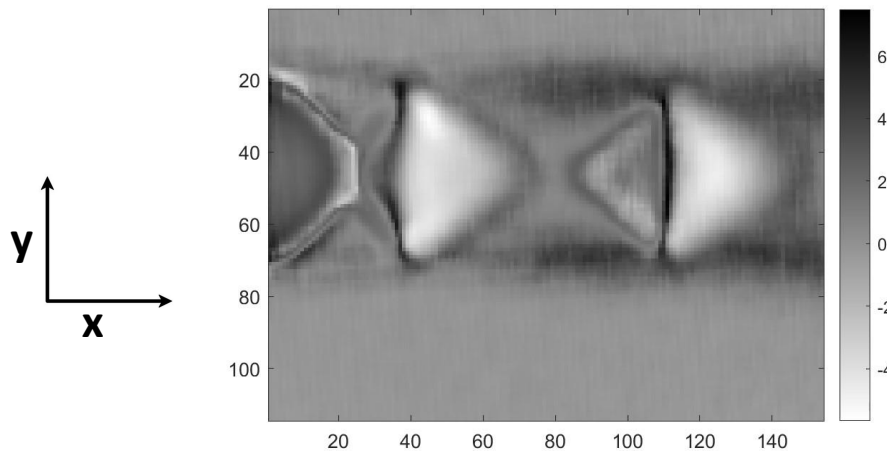


Fig2: Line of sight integrated density gradient (equivalent to Schlieren) obtained from BOS

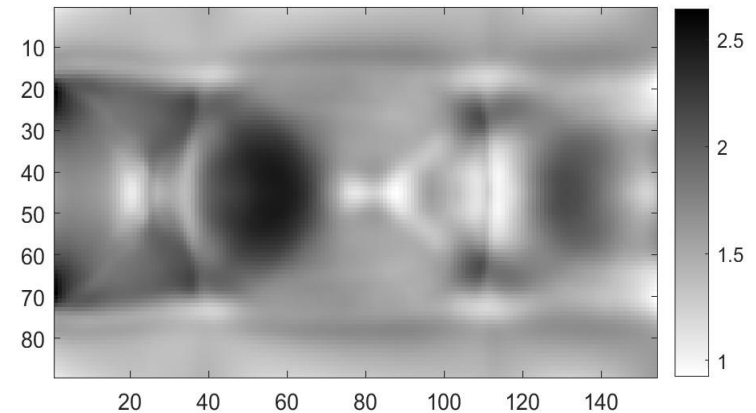


Fig3: Reconstructed density field, from the BOS data using filtered back projection.

Work done during Summer

Determination of the appropriate fin angles

- Mach number of the flow is 2.87
- The RR-MR transition is determined by the von-Neumann and detachment criteria.
- The von-Neumann criterion gives the transition angle from RR to the dual solution domain to be around 19.3 degrees.
- The detachment criterion gives the transition angle from the dual solution domain to MR to be around 20.7 degrees
- The double fin configuration thus is planned to have angles varying from around 18 degrees to 21 degrees, to capture the dual solution domain as well.

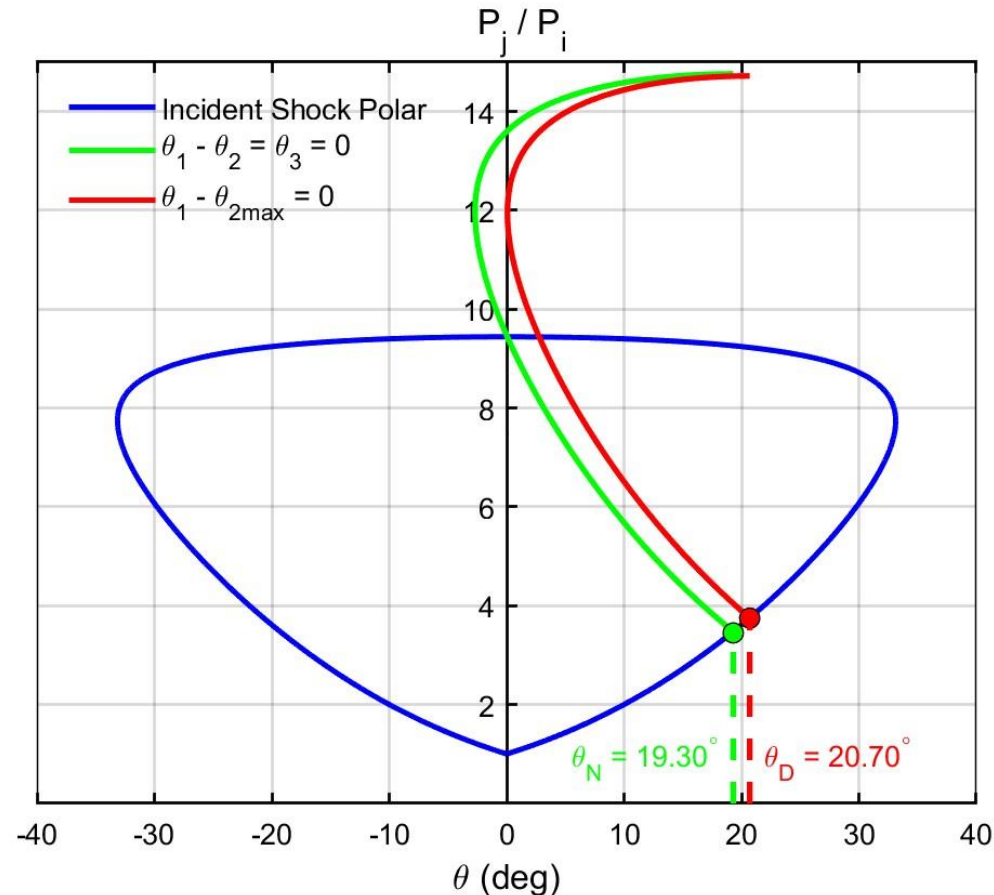


Fig4: Shock polar depicting the von Neumann (θ_N) and detachment (θ_D) criteria transition points for shock interactions.



Project Timeline

Project Timeline

- **September - October:** Design and fabrication of the experimental set-up
- **November - December:** Experiments with conventional schlieren, oil flow and unsteady pressure measurements.
- **January - February :** Implementation of tomo-BOS to obtain 3D visualisation.

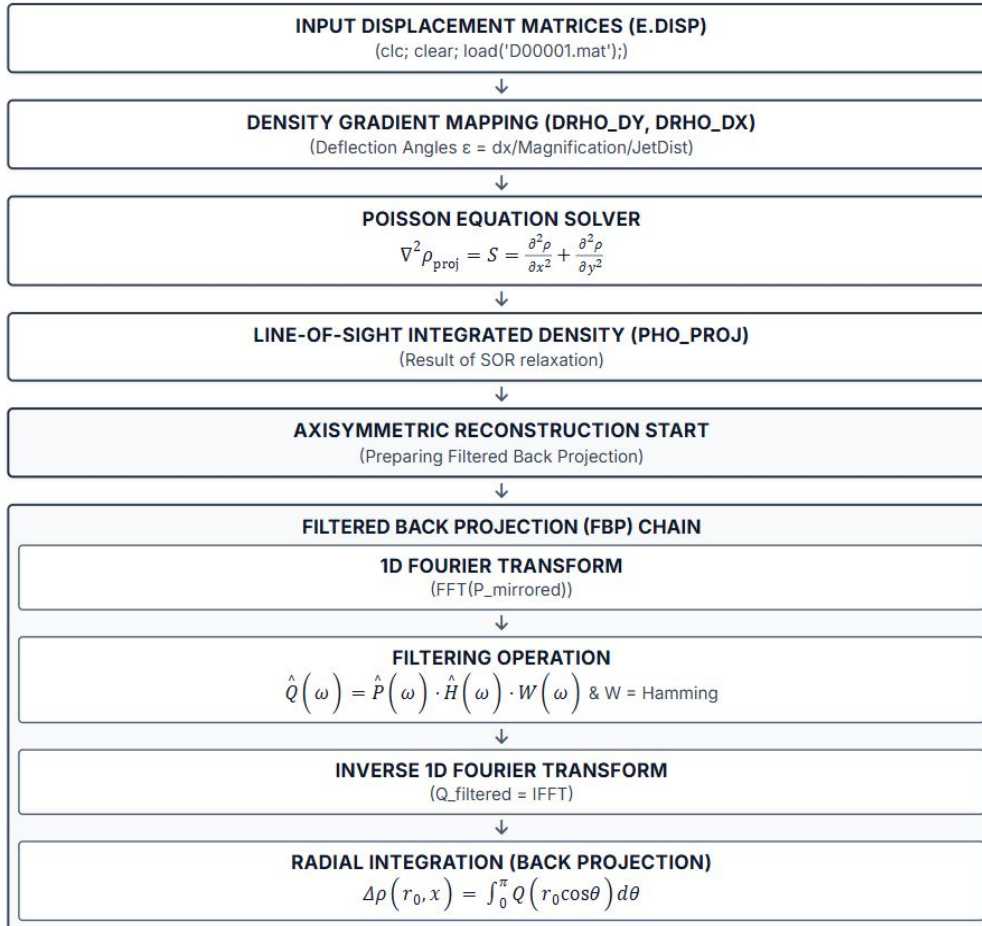


References

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- [2]: Gaitonde, D.V., Adler, M.C., “Dynamics of Three-Dimensional Shock-Wave/Boundary-Layer Interactions”, *Annu. Rev. Fluid Mech.*, **55**: 291-321, 2023.
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- [8]: Seçkin, S., et al., “Surface Properties of Double-Fin Generated Shock-Wave/Boundary-Layer Interactions”, *AIAA J.*, **61**: 4825-4838, 2023.
- [9]: Mercier, B., Lacassagne, T., “Design rules for Background Oriented Schlieren experiments with least-squares based displacement calculation”, *Flow Meas. Instrum.*, **92**: 102395, 2023.
- [10]: Venkatakrishnan, L., Meier, G.E.A., “Density measurements using the Background Oriented Schlieren technique”, *Exp. Fluids*, **37**: 237-247, 2004.

Appendix

Filtered Back Projection:



- 1 **Single Projection Assumption ($\theta = 0$):** Due to jet axisymmetry, all angular projections are identical to $P_{\theta=0}(l) = P(l)$.
- 2 **Filtering for Reduced Noise:** A 1D frequency filter $\hat{H}(\omega)$ is constructed using a Shepp-Logan taper combined with a Hamming window to clip high frequencies and suppress experimental noise.
- 3 **Local Density Reconstruction:** The local radial density distribution $\rho(r)$ is directly recovered from the single-plane projection by evaluating the back-projection integral over the mirrored radial domain.